

Hackathon Problem Statements

Brief Problem Statement Collection — Key Considerations Removed

This document contains the ten problem statements from the uploaded Hackathon Problem Statement Booklet, rewritten briefly in paragraph form. The original problem descriptions are retained in meaning while the Key Considerations sections have been removed.

1. Financial Goal Conflict Detector

Individuals often have multiple financial goals that they want to achieve at different stages of their lives. These goals may include retirement planning, children's education, buying a home, and maintaining an emergency fund. However, people usually have limited income and savings available to meet all these goals. When several goals compete for the same financial resources, it becomes difficult to understand which goals should receive priority. Individuals may unknowingly allocate insufficient funds to important goals. They may also set unrealistic deadlines without understanding whether their income and savings can support them. A lack of systematic financial planning can make these conflicts difficult to identify. There is a need for a system that can analyze multiple financial goals together. The system should help users understand possible conflicts between their goals and available resources. It should provide a clear view of whether their desired goals and timelines are financially achievable.

2. AI Health Report Translator

Medical and laboratory reports contain complex medical terminology that can be difficult for ordinary patients to understand. Reports may include abbreviations, technical descriptions, measurements, and medical findings. Patients without medical knowledge may struggle to understand what these results actually mean. This can create confusion and unnecessary anxiety about their health condition. Many patients depend on informal internet sources or other people to interpret their reports. In some situations, they may have to wait until their next doctor's appointment to understand important information. Misunderstanding a medical report can result in patients ignoring important findings. It may also cause patients to make inappropriate health-related decisions. There is a need for an AI-based system that can translate complex medical information into simpler language. The system should help patients understand their reports while making it clear that professional medical advice is still required for diagnosis and treatment decisions.

3. AI Medical Safety Companion

Patients who take multiple medications may find it difficult to remember the correct dosage and timing of every medicine. Different medicines may have different schedules and instructions that must be followed carefully. Forgetting a dose can affect the effectiveness of treatment. Taking the wrong dosage or taking medicines at the wrong time can also create safety risks. Elderly patients may face additional difficulties when managing several medications. People with chronic health conditions may need to follow medication schedules for long periods. Drug interactions can also become a concern when multiple medicines are taken together. Manual methods such as paper charts and relying only on memory can easily lead to mistakes. There is a need for an intelligent medical safety companion that can assist users in managing their medication schedules. Such a system can help users track medicines, remember dosage timings, and reduce the possibility of medication-related errors.

4. AI Urban Stress Map

Modern cities experience several urban challenges that can affect the quality of life of residents. These challenges include traffic congestion, excessive noise, high temperatures, pollution, and overcrowding. Such problems are often monitored separately using different systems and sources of information. Because the information is divided between different monitoring systems, it can be difficult to understand the overall condition of a particular area. Some locations may experience several stress factors at the same time. City planners may therefore have difficulty identifying which areas require immediate attention. Separate monitoring systems can make it harder to prioritize resources effectively. Without a unified view of urban stress, infrastructure planning may also become less efficient. There is a need for an intelligent system that can combine information from different urban sources. The system should identify areas experiencing multiple stress factors and present them in a unified urban stress map. This can help city planners understand problem areas and support better planning and resource allocation.

5. AI City Problem Predictor

Many problems in cities are addressed only after they have already occurred. This reactive approach can result in delayed responses to infrastructure and public-service issues. When problems are detected late, emergency services and maintenance teams may face additional pressure. Municipal resources and budgets can also be affected by unexpected problems. For example, infrastructure failures may require immediate attention instead of being handled through planned maintenance. Cities generate large amounts of information that can potentially be used to identify patterns before problems occur. However, this information is not always used effectively for predictive planning. There is a need for an AI-based system that can analyze available urban data and identify possible future problems. The system could provide predictive insights to city authorities before certain issues become serious. Early identification can support proactive infrastructure maintenance and planning. It can also help authorities allocate resources more efficiently and improve the overall management of urban services.

6. AI Personalized Learning Platform for Rural Communities

Students in rural communities may have different learning abilities, backgrounds, and levels of understanding. However, many rural schools have limited access to personalized educational resources. Teachers may have to manage large classrooms containing students with different learning levels. Providing individual attention to every student can therefore be difficult. Students who struggle with particular subjects may not receive sufficient personalized support. Existing e-learning platforms may also depend heavily on stable internet connectivity and modern devices. These requirements can make some digital learning solutions impractical for students in rural areas. There is a need for an AI-based learning platform designed specifically to support such environments. The platform should be capable of adapting learning content according to individual student needs and learning levels. It should also consider connectivity and device limitations when delivering educational resources. Such a system could provide personalized learning support while helping teachers manage mixed-ability classrooms more effectively.

7. Learning Debt Detection

Students can sometimes achieve good examination scores even when they have gaps in their understanding of important foundational concepts. Traditional grading systems generally focus on marks and overall academic performance. As a result, hidden weaknesses in prerequisite topics may not always be identified. These unresolved gaps can affect a student's ability to understand more advanced concepts later. Over time, small knowledge gaps can accumulate and become larger learning difficulties. A student may continue progressing through different topics without fully understanding the concepts required for them. Teachers may also find it difficult to manually identify every individual student's learning gaps. There is a need for a system that can detect and track these hidden knowledge gaps. An intelligent learning system could analyze student performance across individual concepts rather than relying only on overall scores. Identifying learning debt early can allow appropriate learning support to be provided before these gaps significantly affect future academic progress.

8. AI-Driven Crop Disease Prediction & Management

Farmers often identify crop diseases only after visible symptoms become noticeable on plants. By the time symptoms appear clearly, the disease may have already caused significant damage. Crop damage can reduce agricultural yield and create financial losses for farmers. Manual inspection of large agricultural fields is also time-consuming and may not detect infections during their early stages. Some infections may not be visually obvious during the initial stages of development. Delayed identification can therefore make disease management more difficult. Farmers may respond by applying pesticides after the disease has already spread. Large-scale reactive pesticide use can increase farming costs and may negatively affect the environment. There is a need for an intelligent system that can identify early signs of crop diseases. The system should help farmers detect infections at an early stage and support timely preventive or management actions.

9. Agricultural Recommendation Conflict Resolver

Farmers receive agricultural recommendations from several different sources when making decisions about their crops. These sources may include weather forecasts, soil monitoring systems, pest alerts, and market information. Sometimes, the recommendations provided by these sources may conflict with one another. Different systems may suggest different actions for irrigation, fertilization, pest management, or harvesting. Farmers may find it difficult to determine which recommendation should be followed. This problem can become more challenging when the information comes from specialized technical systems. Farmers may not always have the technical expertise required to compare and reconcile these recommendations. Following conflicting advice without proper guidance can result in inefficient agricultural decisions. Such decisions may lead to crop damage, resource wastage, or financial losses. There is a need for an intelligent system that can collect recommendations from multiple sources and identify conflicts between them. The system should help farmers understand the conflicting information and provide a consolidated basis for making agricultural decisions.

10. Intelligent Pesticide Sprinkling Based on Plant Infection Level

Farmers commonly apply pesticides across entire agricultural fields even when only certain plants or specific areas are infected. Uniform pesticide application means that healthy plants may also receive unnecessary chemicals. This can result in excessive pesticide consumption and increase the overall cost of farming. Unnecessary pesticide application can also affect beneficial insects and contribute to long-term environmental problems. Applying pesticides uniformly does not account for differences in infection levels between individual plants. Some plants may be healthy, while others may have mild or severe infections. There is therefore a need for a system capable of identifying the infection level of individual plants. The system should analyze plants in real time and determine which plants or areas require treatment. Pesticide application could then be adjusted according to the detected infection severity. Targeted spraying can help reduce unnecessary pesticide usage while focusing treatment on infected plants.